

# Acid–Base Titration Curves

Equivalence points, pH during a titration, and indicators

Name \_\_\_\_\_ Class \_\_\_\_\_ Date \_\_\_\_\_

## INDICATORS (COLOUR-CHANGE RANGE, APPROX.)

Methyl orange: pH 3.1–4.4    Bromothymol blue: pH 6.0–7.6

Phenolphthalein: pH 8.2–10.0

Give pH to two decimal places.

## A. Understanding

1. Explain the difference between the **equivalence point** and the **end point** of a titration.

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2. Predict whether the pH at the equivalence point is below 7, equal to 7, or above 7 for each titration:

(a)  $\text{HNO}_3$  with KOH (b) formic acid ( $\text{HCOOH}$ ) with NaOH (c) HCl with ammonia

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## B. Strong acid with strong base

20.00 mL of 0.150 M HCl is titrated with 0.100 M NaOH.

3. What volume of NaOH is needed to reach the equivalence point?

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4. What is the pH before any NaOH is added?

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5. What is the pH after 10.00 mL of NaOH has been added?

6. What is the pH after 40.00 mL of NaOH has been added?

### C. Weak acids and indicators

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7. During the titration of an unknown weak acid, the pH at the half-equivalence point is 4.20. What are the  $pK_a$  and  $K_a$  of the acid?

8. Which indicator from the data box would you choose for titrating a weak acid with NaOH? Explain your choice.